The finite ("other") terms: verification and a compact relation

1 Verdict of the check

Every row of the uploaded file was checked against the corresponding (previously verified) Group-I/II/III result, demanding

$$\text{(source, exact in } \Lambda, \vee) \;=\; \log\!\left[\tfrac{\vee}{k^+}\right]\times\text{(BFKL row)} + \log\!\left[\tfrac{k^+}{\Lambda}\right]\times\text{(JIMWLK row)} + \text{(this file's row)} + \mathcal{O}(\Lambda).$$

The closed-form rows were checked *analytically* (the log splits below are exact identities, and the kernels and colour brackets agree verbatim); Group I Row III was in addition checked *numerically against the p^+ master integrals* (principal-value quadrature, five random configurations, two values of $\vee$: relative residual $\sim 10^{-6}$, consistent with $\mathcal{O}(\Lambda)$ at $\Lambda = 10^{-6}k^+$).

Row	status	remarks
G-I II (N_1)	✓	bracket $= \frac{11}{3} - 2\log\!\left[\frac{k^+}{\Lambda}\right] + \ell$ exactly; the $-2\log\!\left[\frac{k^+}{\Lambda}\right]$ piece is the (certified) JIMWLK term; the $\frac{11N_c}{3}\left(\frac{2}{\epsilon}+\dots\right)$ piece is silently dropped — see §2(b)
G-I II (N_{2b})	✓	R^i identical to the source; $\vee\to\infty$ finite
G-I III ($N_{1..3b}$)	✓*	exact vs. the p^+ masters, but only with $\log\lvert\cdot\rvert$ (PV) — see §2(a)
G-I IV	✓	exact ($\mathbb{L}$- and ℓ-remnants of the verified source)
G-I V (both parts)	✓	the Group file's own $[\,]_+$ representation with the UV piece split off; $\vee$-finite; $[\,]_+$ undefined, fate of the $\frac{11}{6}$ UV block unstated — §2(b)
G-II I (N_{1a}, N_{1b})	✓	exact
G-II I (N_{2a}–N_{2f})	✓	exact (N_{2a} is *entirely* finite: $\vee$ cancels between the two logs); underlying p^+ integration re-verified to 10^{-22}; the three N_{2f} pieces are integrable only as a sum
G-II IV	✓	matches the corrected P_{gg}/subtraction construction; $\mathcal{P}_{gg}$ undefined in the file; $2/\epsilon$ retained by design (§2(b))
G-III I, II	✓	exact
G-III III, V, VI	✓	the two nontrivial rearrangements verified analytically (§5): the $[\,]_+$ channel shift produces exactly $+\log\!\left[\frac{\vee}{k^+}\right]$, and the $y'\to z$ recoil subtraction integrates to exactly $\int_0^{\vee-k^+}\frac{dp^+}{p^++k^+} = \log\!\left[\frac{\vee}{k^+}\right]$, both reappearing in the BFKL file (its G-III entries) with the right kernels, so no double counting

2 Two fixes required

(a) Restore the absolute values (PV) in Group I Row III. This revision replaced the earlier $\log|\cdot|$ by plain $\log[\cdot]$. The arguments

$$\frac{(x-w)^2-(x-z)^2}{(x-w)^2}, \qquad \frac{\frac{k^+}{\vee}(x-w)^2-(x-z)^2}{(x-w)^2-(x-z)^2}, \qquad \frac{(x-z)^2-\frac{k^+}{\vee}(x-w)^2}{(x-z)^2-(x-w)^2}$$

change sign on half of the (x,z,w) phase space, so as written these terms are complex. Numerically: with $\log|\cdot|$ the Row-III identity against the p^+ masters holds to 10^{-6}; with plain logs it fails at $\mathcal{O}(0.1 \div 0.4)$ with a spurious imaginary part. The $|\cdot|$ is exactly the PV reading of the p^+ poles, and after the collapse of §4.2 only *one* absolute value survives, in $\log\big|(x-z)^2-\bar{\xi}\,(x-w)^2\big|$.

(b) Restore the bookkeeping statements that were deleted. (i) definitions of $[1/p^+]_+$, $[k^+/p^+]_+$, $[k^+/(p^+(p^+ + k^+))]_+$ (subtraction at $p^+ = 0$ on the stated interval) and of $\mathcal{P}_{gg}(\xi)$; (ii) a sentence stating that the $\frac{11N_c}{3}\big(\frac{2}{\epsilon}+2\gamma_E+\log\frac{(x-w)^2\mu^2}{4}\big)$ piece of G-I Row II N_1 and the $\frac{11}{6}$ UV block of G-I Row V N_{2b} (commented out) are moved to the renormalization class of §4.4, where together with the β_0-term of G-II Row IV they build the running coupling, $\beta_0 = \frac{11N_c}{3}$.

3 The compact relation

Let $\bar{\xi} \equiv k^+/\vee$, and

$$\ell \equiv \log(1-\bar{\xi}), \qquad u \equiv \frac{(x-z)^2}{(x-w)^2}, \qquad \mathbb{L}(A,B) \equiv \log\Big[\frac{(1-\bar{\xi})A+\bar{\xi}B}{B}\Big], \qquad \mathcal{L} \equiv \log\big|u-\bar{\xi}\big|,$$

$$\mathcal{A}_0 = \frac{\theta_w}{\big|(x-w)\wedge(x-z)\big|}, \qquad \theta_w = \angle\big(x-w,\ z-w\big).$$

Then the complete NLO cross-section organizes as

$$\boxed{\frac{d^3N_{\rm NLO}}{d^3k} = \log\big[\tfrac{\vee}{k^+}\big]\Big[K_{\rm BFKL}\otimes\frac{d^3N_{\rm LO}}{d^3k}\Big] - \log\big[\tfrac{k^+}{\Lambda}\big]\Big[H_{\rm JIMWLK}\,\frac{d^3N_{\rm LO}}{d^3k}\Big] + \frac{d^3N_\delta}{d^3k}\ ,} \tag{1}$$

(the first two brackets verified earlier at operator level: the three- and four-ρ logarithms cancel identically and the two-ρ $\log\big[\frac{k^+}{\Lambda}\big]$ set equals $-H_{\rm JIMWLK}N_{\rm LO}$ exactly), and the finite part falls into *five* classes,

$$\boxed{\frac{d^3N_\delta}{d^3k} = \underbrace{N_\delta^{\rm UV}}_{\beta_0,\,\mathcal{P}_{gg}} + \underbrace{N_\delta^{\partial}}_{\propto\ell} + \underbrace{N_\delta^{\rm coll}}_{\propto\mathbb{L}} + \underbrace{N_\delta^{\rm geo}}_{\mathcal{L},\,\mathcal{A}_0,\,\log\frac{(z-w)^2}{(x-w)^2}} + \underbrace{N_\delta^{\rm rec}}_{[\,]_+\text{-convolutions}}} \tag{2}$$

Operationally, the whole file is the image of the Group-file results under the exact substitution table ($\Lambda \to 0$ at fixed $\bar{\xi}$; every line is an identity):

source structure (exact in $\Lambda, \vee$)	evolution part	finite remnant
$\log(\frac{\vee}{k^+} - 1)$	$\log\big[\frac{\vee}{k^+}\big]$	ℓ
$\log \frac{(\vee - k^+)A + k^+ B}{\Lambda A + k^+ B}$	$\log\big[\frac{\vee}{k^+}\big]$	$\mathbb{L}(A, B)$ (argument > 0, no PV)
$\log \frac{\vee}{\Lambda + k^+}$	$\log\big[\frac{\vee}{k^+}\big]$	0
$\log \frac{k^+ - \Lambda}{\Lambda}$	$\log\big[\frac{k^+}{\Lambda}\big]$	0
$\frac{1}{\Lambda} \pm \frac{1}{k^+ - \Lambda}$, $\frac{1}{\Lambda} - \frac{1}{\vee - k^+}$	—	cancel pairwise
$\mathcal{A}$(arctan)	—	$\mathcal{A}_0 = \theta_w / \lvert p \wedge q \rvert$
$\log \frac{k^+ B - \vee A}{k^+ B - (k^+ + \Lambda)A}$	$\log\big[\frac{\vee}{k^+}\big]$	$\log\big\lvert\frac{\bar\xi - u}{1 - u}\big\rvert$ (PV)
$\log \frac{k^+ B - (k^+ - \Lambda)A}{k^+ B - \Lambda A}$	0	$\log \lvert 1 - u \rvert$ (PV)
$\log \frac{\vee A - k^+ B}{(\Lambda + k^+)A - k^+ B}$	$\log\big[\frac{\vee}{k^+}\big]$	$\log\big\lvert\frac{u - \bar\xi}{u - 1}\big\rvert$ (PV)
$\log \frac{[k^+(x-w) - (k^+ - \Lambda)(x-z)]^2}{[k^+(x-w) - \Lambda(x-z)]^2}$	0	$\log \frac{(z-w)^2}{(x-w)^2}$
$\frac{2}{\epsilon} + 2\gamma_E + \log \frac{r^2 \bar\mu^2}{4}$	—	$\to$ class N_δ^{UV}
$\int_\Lambda^{\vee - k^+} \frac{dp^+}{p^+} f(p^+)$	$(\log\big[\frac{k^+}{\Lambda}\big] + \log\big[\frac{\vee}{k^+}\big] + \ell) f(0)$	$\int_0^{\vee - k^+} dp^+ \big[\frac{1}{p^+}\big]_+ f$ ($\vee$-finite)

Here $A = (x-z)^2$, $B = (x-w)^2$ (and primed copies for the conjugate side), $u = A/B$.

4 The simplified (collapsed) forms

4.1 The universal $\mathbb{L}$-family (G-I IV; G-II I; G-III I, II)

All these rows are built from *two* universal rank-2 kernels,

$$T^{ij}(x; w, z) = \frac{(x-z)^j (z-w)^i}{(z-w)^2 (x-z)^2} - \frac{(x-w)^i (z-w)^j}{(z-w)^2 (x-w)^2} - \frac{(x-z)^j (x-w)^i}{(x-z)^2 (x-w)^2} + \frac{\delta^{ij}}{2(z-w)^2}, \tag{3}$$

$$S^{ij}(x; w, z) = \frac{(x-w)^i (z-w)^j}{(z-w)^2 (x-w)^2} + \frac{(x-w)^i (x-z)^j}{2(x-w)^2 (x-z)^2}, \tag{4}$$

and each such row reads, with its own legs and colour bracket $\mathfrak{B}_r$,

$$\boxed{N_\delta^r = c_r \, \mathcal{K}^i(a_r) \, \mathcal{K}^j(b_r) \Big[\mathbb{L}\big((x-z)^2, (x-w_r)^2\big) \, T^{ij}(x; w_r, z) + \ell \, S^{ij}(x; w_r, z) \Big] \otimes \mathfrak{B}_r} \tag{5}$$

(r = G-I IV: $c = \frac{ig^4}{8\pi^5 k^+}$, legs $(x'-w'), (y-z)$; G-II I*a*: $c = -\frac{ig^4}{16\pi^5 k^+}$, same legs; G-II I*b*: $c = +\frac{ig^4}{16\pi^5 k^+}$, legs $(x'-w), (y-z)$, $w \to w'$ inside; G-III I, II: $c = -\frac{ig^4}{8\pi^5 k^+}$). The G-II Row I second part (N_{2a}–N_{2f}) is one single colour structure

$$\mathfrak{G} = f^{c'bd'} f^{cbd} U^{ac'}(w') U^{ac}(w) \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'} f^{acd} U^{ac'}(w') U^{cb}(z) U^{de}(x) \rho^{d'}(x') \rho^{e}(x) - (\,' \leftrightarrow\,) + N_c U^{de'}(x') U^{de}(x) \rho$$

multiplied by rational coefficients times only $\mathbb{L}\big((x-z)^2, (x-w)^2\big)$, $\mathbb{L}\big((x'-z)^2, (x'-w')^2\big)$, ℓ, $\big[\frac{1}{\vee} - \frac{1}{k^+}\big]$, and the entirely finite $\log \frac{(1-\bar\xi)(x-z)^2 + \bar\xi (x-w)^2}{\bar\xi (x'-w')^2 + (1-\bar\xi)(x'-z)^2} - \log \frac{(x-w)^2}{(x'-w')^2}$ of N_{2a} (in which $\vee$ cancels identically).

4.2 Group I Row III collapses to four structures

Write the common bracket $\mathrm{Br} = \big[U^{ab'}(x') - U^{ab'}(w')\big] \rho^{b'}(x') \big[U^{de}(x) U^{bc}(z) - U^{de}(x) U^{bc}(y)\big] \rho^{e}(x) \rho^{c}(y)$ and $E' = (x'-w')^2$, $F = (y-z)^2$, $A = (x-z)^2$, $B = (x-w)^2$, $C = (z-w)^2$, $G = (x-w)\cdot(x-z)$. Using $\log\big\lvert\frac{u-\bar\xi}{u-1}\big\rvert + \log\lvert 1-u\rvert = \mathcal{L}$ in N_1, and the fact that the six $\log\big\lvert\frac{\bar\xi - u}{1-u}\big\rvert$ kernels of N_2 and the six $\log\lvert 1-u\rvert$ kernels of N_{3a} are *the same set* Σ (so all $\log\lvert 1-u\rvert$ cancels), the 23 finite terms of Row III

reduce *exactly* (checked numerically to 10^{-24}) to

$$\boxed{N_\delta^{\rm RIII} = \frac{1}{(2\pi)^3}\frac{ig^4 f^{abd}}{16\pi^5 k^+}\int e^{-ik(w'-w)}\Big\{\big(Y-2K_2+2\Sigma\big)\,\mathcal{L}-\big(Y+2X_{12}\big)\,\ell+2\,\mathcal{A}_0\,W+\Xi\,\log\tfrac{C}{B}\Big\}\otimes \mathrm{Br}} \tag{6}$$

with $\mathcal{L}=\log\big|u-\bar\xi\big|=\log\Big|\frac{(x-z)^2-\bar\xi\,(x-w)^2}{(x-w)^2}\Big|$ and

$$Y=\frac{(x'-w')\cdot(y-z)}{E'F}\,\frac{A+B}{(A-B)^2},\qquad K_2=\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{E'BAF},$$

$$X_{12}=\frac{(x'-w')\cdot(y-z)\;(z-w)\cdot\big[B(x-z)-A(x-w)\big]}{E'FC\,(B-A)^2},$$

$$\Sigma=X_{12}+\frac{1}{E'FC(B-A)}\Big[\frac{A}{B}(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)+\frac{B}{A}(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)$$
$$-(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)-(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)\Big],$$

$$\Xi=\frac{1}{E'FC}\Big[\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{B}-\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{A}-(x'-w')\cdot(y-z)\Big].$$

The $\mathcal{A}_0$ coefficient has the manifestly finite wedge form (verified exactly):

$$\mathcal{A}_0\,W=\frac{\theta_w\;\mathrm{sgn}\big[(x-z)\wedge(x-w)\big]}{E'F\,C}\left[\frac{(x'-w')\cdot(z-w)\,\big[(x-z)\wedge(y-z)\big]}{(x-z)^2}-\frac{(y-z)\cdot(z-w)\,\big[(x-w)\wedge(x'-w')\big]}{(x-w)^2}\right] \tag{7}$$

i.e. the arctangent $\mathcal{A}_0$ enters only through the bounded angle θ_w — no denominator $\sqrt{AB-G^2}$ survives. Moreover the coefficient $Y-2K_2+2\Sigma$ of $\mathcal{L}$ is *regular* at $u=1$ (the $1/(A-B)^2$ and $1/(A-B)$ poles cancel identically between Y, $2X_{12}$ and the rest of Σ), so after the collapse the only non-analyticity of Row III is the single PV log $\mathcal{L}$, whose argument vanishes only in the shrinking region $(x-z)^2<\bar\xi\,(x-w)^2$.

4.3 The recoil class (G-I V; G-III III, V, VI)

These rows keep one exact p^+ convolution with the recoil phase $e^{-ik\cdot(z-x)\,p^+/k^+}$; they are finite at $\vee\to\infty$ by oscillation. Two exact identities fix their bookkeeping against the BFKL file:

$$\int_0^P dp^+\Big[\tfrac{1}{p^+}\Big]_+ g=\int_0^P dp^+\Big(\Big[\tfrac{k^+}{p^+(p^++k^+)}\Big]_++\tfrac{1}{p^++k^+}\Big)g\;-\;g(0)\,\log\big[\tfrac{\vee}{k^+}\big],\qquad P=\vee-k^+, \tag{8}$$

$$\int_0^{\vee-k^+}\frac{dp^+}{p^++k^+}=\log\big[\tfrac{\vee}{k^+}\big]\qquad(\text{the } y'\to z \text{ subtraction term}), \tag{9}$$

and $\int_{y'}e^{-ik(y'-w)}\,\mathcal{K}^m(y'-z)\big|_{\rm FT}=\int_{w'}e^{-ik(w'-w)}\mathcal{K}^m(w'-z)$ reproduces exactly the $U(z)$-terms with $(w'-z)$ kernels in the BFKL file's Group-III entries. The two induced $\log\big[\frac{\vee}{k^+}\big]$ pieces appear there with opposite signs against the source's blue log — the net BFKL content of these rows is the recoil-subtraction term alone, and nothing is double counted.

4.4 The UV / collinear class

Collecting every $\big(\frac{2}{\epsilon}+2\gamma_E+\log\frac{r^2\bar\mu^2}{4}\big)$: the $\mathcal{P}_{gg}$ convolution of G-II Row IV (collinear splitting of the measured gluon, with its $(1-\xi)/\xi$ real-emission subtraction and the $(1-\bar\xi)$ boundary term), the β_0 term of G-II Row IV, the $\frac{11N_c}{3}$ piece of G-I Row II and the $2\times\frac{11}{6}N_c$ piece of G-I Row V combine with $\beta_0=\frac{11N_c}{3}$ into the one-loop renormalization of g^2 in $d^3N_{\rm LO}$ plus the standard $\mathcal{P}_{gg}\otimes\log$ collinear remainder; the $\ell\times(\frac{2}{\epsilon}+\dots)$ piece of G-I Row II is the $\int_{\bar\xi}^1 d\xi\,\frac{2N_c\xi}{1-\xi}$ endpoint of the same convolution. (This assembly is structural; the final coefficient check should be done once the $\mathcal{P}_{gg}$ and $[\,]_+$ conventions are restored in the file.)

5 The $\vee \to \infty$ limit

As $\bar{\xi} \to 0$: $\ell \to 0$, $\mathbb{L}(A, B) \to \log \frac{A}{B}$, $\mathcal{L} \to \log u$, and the entire closed-form finite sector is generated by just

$$\log \frac{(x-z)^2}{(x-w)^2}, \qquad \theta_w \text{ (via } \mathcal{A}_0), \qquad \log \frac{(z-w)^2}{(x-w)^2},$$

plus the $\vee$-finite recoil convolutions and the $\mathcal{P}_{gg}/\beta_0$ class — every $\vee$-dependence of d^3N_δ is a positive power of $\bar{\xi} = k^+/\vee$.